

Task: ML Experiment Tracking with MLflow

Objective

In this task, you will use **MLflow** to track and compare different machine learning experiments.

You will train the same machine learning model using different hyperparameters and use MLflow to record:

- Parameters
- Metrics
- Model artifacts
- Experiment results

Finally, you will identify the best-performing model.

Scenario

You are working on a **House Price Prediction** problem.

Your goal is to train a regression model and determine which set of hyperparameters produces the best results.

Part 1 — Prepare the Dataset

Use the **California Housing dataset**.

1. Load the dataset.
 2. Split it into:
 - **80% Training**
 - **20% Validation**
 3. Make the split reproducible by using a fixed random seed.
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Part 2 — Train Multiple Experiments

Train the same model **three times** using different hyperparameters.

Use the following configurations:

Experiment Max Depth Learning Rate

Run 1	3	0.1
Run 2	5	0.05
Run 3	7	0.01

For every run, use MLflow to track:

Parameters

- max_depth
- learning_rate

Metrics

- RMSE
- MAE
- R^2

Part 3 — Log the Model

For every MLflow run:

1. Train the model.
2. Calculate the validation metrics.
3. Log the parameters.
4. Log the metrics.
5. Save the trained model as an MLflow artifact.

Your MLflow experiment should allow you to answer:

"Which model configuration produced the best result?"

Part 4 — Compare the Experiments

Open the MLflow UI and compare the three runs.

Create a table similar to:

Run	Max Depth	Learning Rate	RMSE	MAE	R ²
Run 1	3	0.1	...	...	...
Run 2	5	0.05	...	...	...
Run 3	7	0.01	...	...	...

Determine which run is the **best model**.

For this task, use **RMSE** as the primary metric for model selection.

Deliverables

Submit a GitHub repository containing:

mlflow-task/

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|— train.py

|— requirements.txt

|— README.md

train.py

Your script should:

- Load the dataset.
- Split the data.
- Train the three experiments.

- Log parameters and metrics using MLflow.
- Log the trained models.

README.md

Include:

1. A short description of the task.
 2. How to run the project.
 3. Screenshot of the MLflow experiment results.
 4. Comparison table of the three runs.
 5. The selected best model.
 6. A short explanation of **why you selected it**.
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Expected Learning Outcome

By completing this task, you should understand how to:

Train → Track → Compare → Select

an ML model using MLflow.